DATA ENGINEERING CASE STUDY

Calibo AI Academy

Hospital Patient Care Analytics Pipeline — use case, solution, approach and expected outcome

STUDENT NAME
M.V Ashika Reddy

ID NUMBER
2300030398

USE CASE TITLE

Smart Hospital Patient Care Analytics: Predicting High-Risk Patients and Handling Emergencies with Zero-Delay Rescheduling

A data-driven system that collects patient data, finds high-risk patients early, and makes sure that when an emergency arrives, patients who already have appointments are moved to another doctor at the same time — with a confirmation message — so nobody is made to wait.

1 The Problem in Simple Words

📁 **Data is scattered**
Registration, appointments, lab reports, wearable devices and doctor notes live in different systems.

⏳ **Patients wait**
When an emergency comes, doctors leave their OPD patients, and booked patients are delayed or forgotten.

⚠️ **Risk is found late**
Patients with dangerous vitals are not identified early, so care is not prioritised in time.

2 Our Solution: A Smart Control Room

Think of it as a control room that collects data, checks it, scores each patient, manages the doctor calendar and reacts instantly to emergencies. The solution is a six-stage data pipeline:

STAGE 1
Collect
Load patient, appointment, vitals and note data (CSV)

STAGE 2
Check
Find duplicates, missing values, double bookings

STAGE 3
Score Risk
1 point for every danger sign in the vitals

STAGE 4
Schedule
Lock all advance appointments in a doctor calendar

STAGE 5
Emergency
Treat at once, move booked patients to another doctor

STAGE 6
Confirm & Report
Send messages, draw charts, save reports

Example: how one emergency is handled

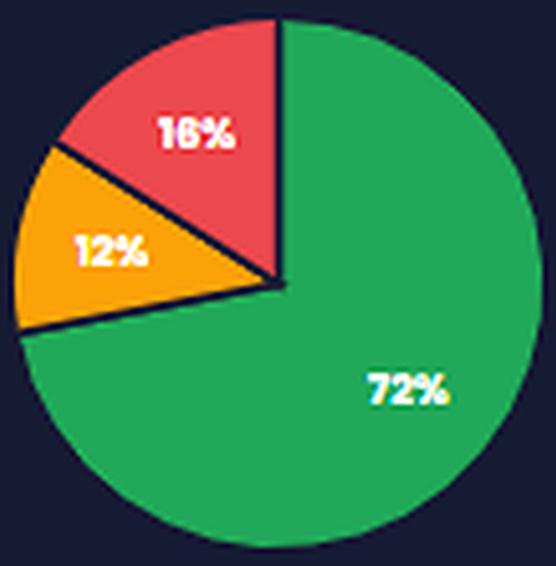
1. At 10:15 a patient arrives in Cardiology with severe chest pain and dangerous vitals.
2. The senior cardiologist takes the emergency **immediately** (0 minutes waiting).
3. Two patients booked with that doctor at 10:45 and 11:00 are moved to another cardiologist **at the same times.**
4. Each moved patient receives an **SMS / WhatsApp confirmation** with the new doctor and room.

3 Understanding the Data (Two Pie Charts)

These charts come from our working model built on 100 patients: 88 with advance appointments and 12 emergency arrivals.

Chart 1 — Patient Risk Level

Who needs the most attention?



Low risk	72 patients
Medium risk	12 patients
High risk	16 patients

Total: 100 patients

What it tells us: 16 out of 100 patients are high risk. The system flags them so a nurse can call them before the visit.

Chart 2 — Effect of Emergencies on Booked Patients

Were booked patients disturbed?



Kept original doctor & time	65 patients
Moved to another doctor (same time)	23 patients

Total: 88 booked patients

What it tells us: 23 of 88 booked patients had to change doctor, but all kept the same time — zero delay.

4 Procedure I Followed (Approach)

1. **Load and understand the data**
   Read the 100-patient CSV: patient details, vitals from wearables, lab status, doctor notes, booked doctor and time.
2. **Check data quality**
   Look for duplicate patients, missing values and double-booked doctors before trusting the data.
3. **Calculate a risk score**
   Give 1 point for each danger sign (abnormal heart rate, low oxygen, abnormal BP, fever, abnormal breathing, not fully alert). 3+ points = High risk.
4. **Build the appointment calendar**
   Place the 88 advance appointments in a schedule table: doctor, date and 15-minute slot.
5. **Handle emergencies in order of arrival**
   Choose the senior doctor of that department, block their time, and move affected patients to another doctor at the same time.
6. **Send confirmations**
   Create a confirmation message for every moved patient (SMS / WhatsApp in a real hospital).
7. **Validate, visualise and save**
   Check there is no delay or double booking, draw pie / bar / line charts and save the final schedule.

Python · Pandas · Matplotlib · Jupyter Notebook · scikit-learn (risk model)

Three golden rules for every emergency

1. See the patient at once
Emergency patients are never made to wait.

2. Same time, another doctor
Booked patients move to a free doctor in the same department (or General Medicine) at the same time.

3. Always confirm
Every moved patient gets a message. If no doctor is free, the patient is flagged for a personal call.

5 Expected Outcome

Results achieved on the sample data

12	23	0 min	0	24
Emergencies handled	Booked patients moved	Maximum delay	Double bookings	Confirmation messages

Expected benefits for the hospital

- ✓ **Shorter waiting times:** booked patients keep their slot even when an emergency happens.
- ✓ **Faster emergency care:** the right specialist is assigned at the moment of arrival.
- ✓ **Early risk detection:** high-risk patients are identified before the visit and get priority care.
- ✓ **Less manual work:** automatic re-scheduling and messages replace phone calls and paper lists.
- ✓ **Better decisions:** charts and reports show department load, doctor workload and risk trends.

Note: the results above are from a 100-record synthetic (fictional) data set used to demonstrate the workflow, not real patient data.

6 Data Engineering Concepts Used

Concept	How it is used in this project
Data sources & ingestion	Registration, appointments, lab status, wearables and notes combined into one input; in production via batch loads and streaming (Kafka).
Data quality	Automatic checks for duplicates, missing values and double bookings before processing.
Transformation	Cleaning, time conversion, risk scoring and triage levels.
Storage layers	Raw → cleaned → final (gold) tables and a SQL warehouse for reports.
Orchestration	Ordered pipeline stages; scheduled with a tool such as Apache Airflow, with emergencies handled as instant events.
Security & privacy	Masked phone numbers and hashed patient keys; personal data kept out of analytics tables.
Serving & analytics	Pie, bar and line charts, final schedule and patient confirmations.

7 Conclusion

This use case shows how a well-designed data pipeline turns scattered hospital data into **faster emergency response, protected appointments and earlier attention for high-risk patients**. With a simple, transparent set of rules, no patient is made late because of an emergency, and every change is confirmed and recorded. The same design can scale to a real hospital by connecting live systems and adding streaming, a data warehouse and monitoring.

SUBMITTED BY

M.V Ashika Reddy

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